

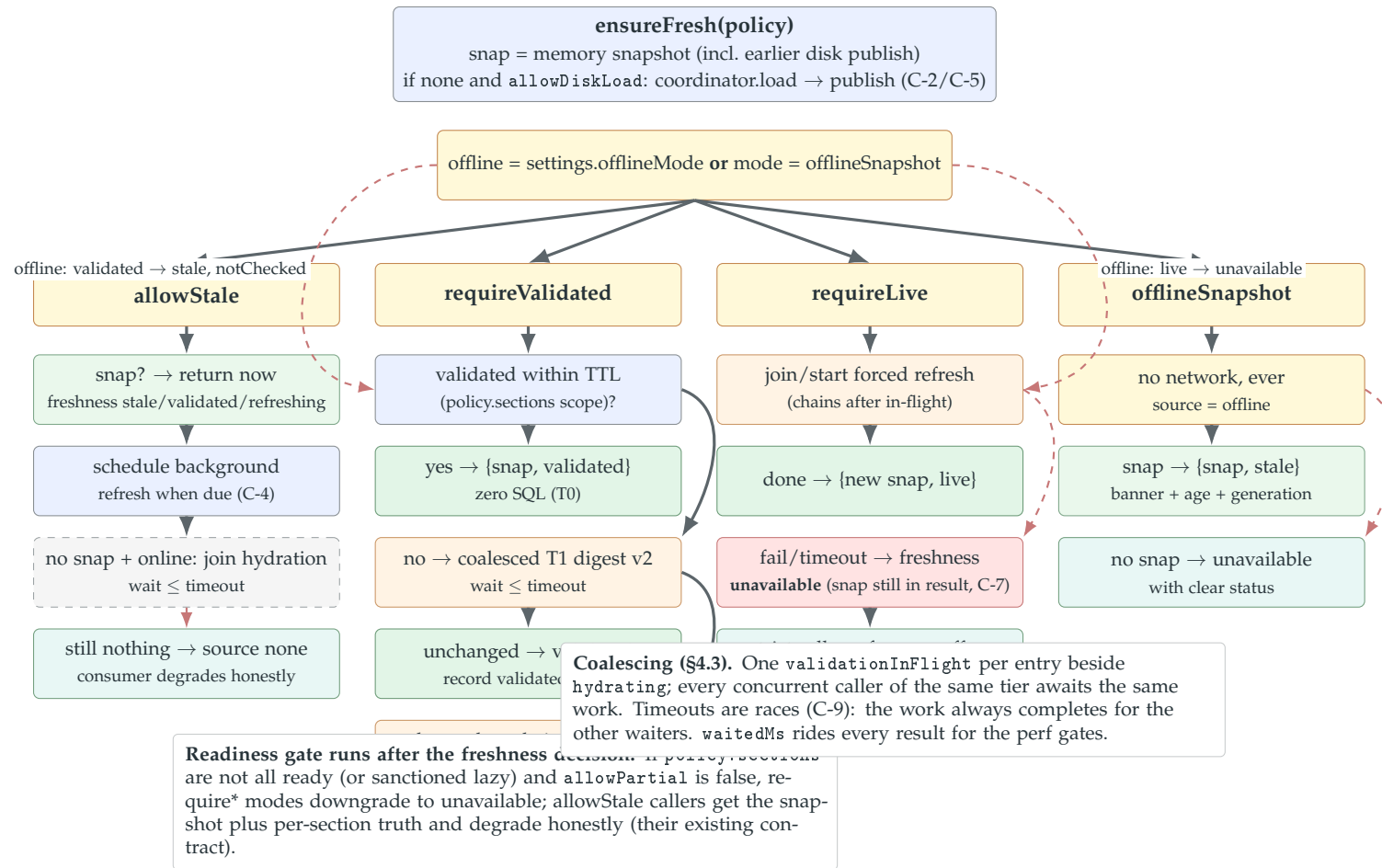
A1. ensureFresh(policy) - the decision procedure

Addendum §4.2 as a flowchart. Waits are races: a timeout stops waiting, it never cancels the shared lane work. Implement this page 1:1.

Metadata substrate - review visuals v2 addendum

companion to REVIEW_ADDENDUM.md - 2026-07-06

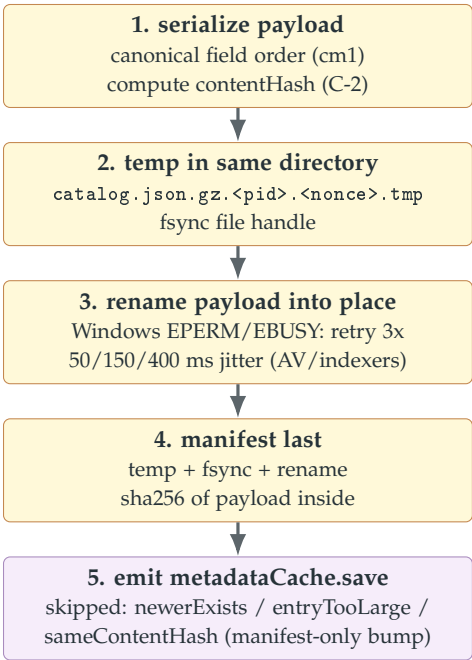
store/registry pure snapshot engine/hydration sessions/data plane contract/privacy gate proposed cache correction/hazard observability



A2. Disk protocol: atomic writes, torn-write recovery, two windows

Every readable state is old-valid or new-valid by construction. The reader trusts nothing the manifest's SHA does not vouch for.

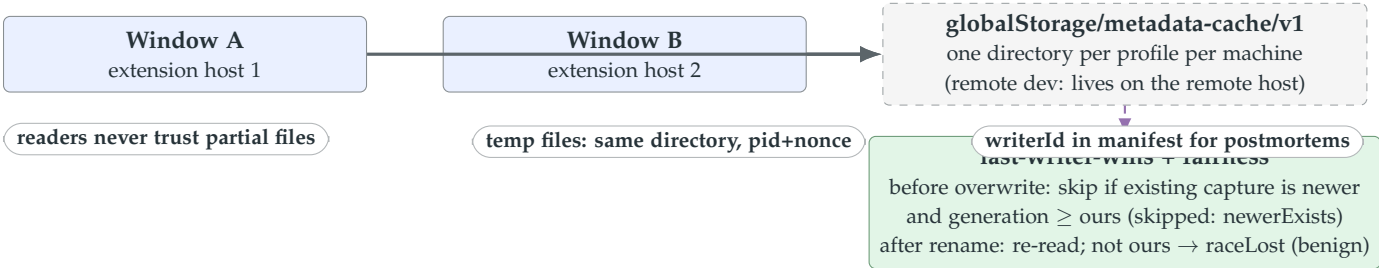
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Torn-write matrix - reader outcome (crash between any two steps)

	manifest: old	manifest: new
payload: old	normal: loads old	unreachable (manifest is written last); defense-in-depth: SHA mismatch → corrupt-safe
payload: new	crash between steps 3–4: manifest SHA no longer matches the bytes → treat corrupt, delete pair, hydrate live (safe; old cache lost)	normal: loads new

Read protocol (base §9.2 + addendum). manifest → validate formatVersion/codecs/key/privacy → payload SHA → shape/counts → policy intersection (C-5: excluded sections become **absent**) → adopt arrays verbatim (§6.1) → publish with source=disk. Any failure: emit corrupt/miss with safe fields, quarantine or delete, continue to live hydration, never crash activation. Recovery budget: p95 < 20 ms before live hydration starts.

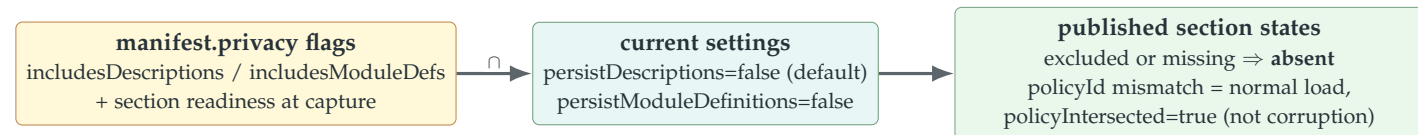


Corrections C-3/C-5/C-6/C-7 on one page. Readiness answers “is this section trustworthy in this snapshot”; freshness answers “is this snapshot recent enough for this caller”. Never collapse them.

Metadata substrate - review visuals v2 **addendum**
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store/registry observability pure snapshot engine/hydration sessions/data plane contract/privacy gate proposed cache correction/hazard

readiness ↓ / source →	live (memory)	disk, not validated	disk, (TTL/T1)	validated	offline mode
ready	serve; freshness live/validated	serve to allowStale; freshness stale	serve to requireValidated		serve with banner + age
failed	render failure; prior snapshot may remain (C-7)	refresh failed: keep disk gen, validation.result=failed	validation says changed but refresh failed → stale, never validated		failed sections stay failed offline
absent	not hydrated yet	excluded by policy at save/load (C-5) → absent, never ready-empty	same		same
stale (reserved)	C-3: means “re-hydration in flight over an existing snapshot” - a transient. Age-based staleness never appears here.				



```
definitions stay lazy
module text is never in payload v1; provider seam keeps definitions:  "lazy"; c
```

failure is never emptiness - now also from disk

FreshCatalogResult.freshness (C-8): **live** = produced by a refresh completed for this call; **validated** = TTL/digest-confirmed; **refreshing** = returned early while shared work continues (allowStale only); **stale** = known-unvalidated; **unfreshness: two axes, no overloaded bit** t met (snapshot may still ride in the result, C-7). **contentHash (C-2)** is the cross-process determinism key; generations stay process-scoped integers.

store/registry observability

pure snapshot

engine/hydration

sessions/data plane

contract/privacy gate

proposed cache

correction/hazard

detects

heuristic / partial

blind

– not applicable

change class	A sniff (local)	B digest v1	B digest v2	TTL (T0)	T2 section (future)	C explicit
CREATE/DROP object - in this editor	detects	detects	detects	–	detects	detects
CREATE/DROP object - external client	blind	detects	detects	–	detects	detects
ALTER object (modify_date bumps) - external	blind	detects	detects	–	detects	detects
sp_rename - external client	blind	blind	detects	–	detects	detects
sp_rename - in this editor	detects	blind	detects	–	detects	detects
ALTER SCHEMA ... TRANSFER - external	blind	blind	detects	–	detects	detects
Column/key/FK/param drift w/o modify_date bump	blind	blind	blind	–	detects	detects
Permission change (section visibility)	blind	heuristic / partial	heuristic / partial	–	heuristic / partial	detects
Database renamed under the lease (H-5)	blind	blind	detects (DB_NAME probe)	–	–	detects
Dynamic SQL DDL via EXEC - in this editor	heuristic / partial → digest	detects	detects	–	detects	detects
Nothing changed (cost of a false alarm)	–	heuristic / partial checksum collision	heuristic / partial checksum collision	detects zero SQL	heuristic / partial	–

digest v2 (H-1 + H-5, one scan, same cost)
SELECT DB_NAME(), COUNT(*),
CHECKSUM_AGG(CHECKSUM(object_id, schema_id,
\\ CAST(name AS varbinary(256)), modify_date))
name as varbinary = byte-exact; plain CHECKSUM over char is
collation-folded (pure-case rename on CI server would still hide)

coverage honesty (unchanged wording)
cheap digest says “likely unchanged”, never proof. It cannot
see column/key/FK/param/description drift or permission
changes. Strict operations still validate or refresh the facts
they script. T2 section digests are the eventual answer, not a
bigger checksum.

poll governance (H-3)
suspend on window blur > 2 min
backoff 60 → 120 → 300 s, jitter ±10%
serverless starts at cap (auto-pause)
store-wide validation semaphore = 2

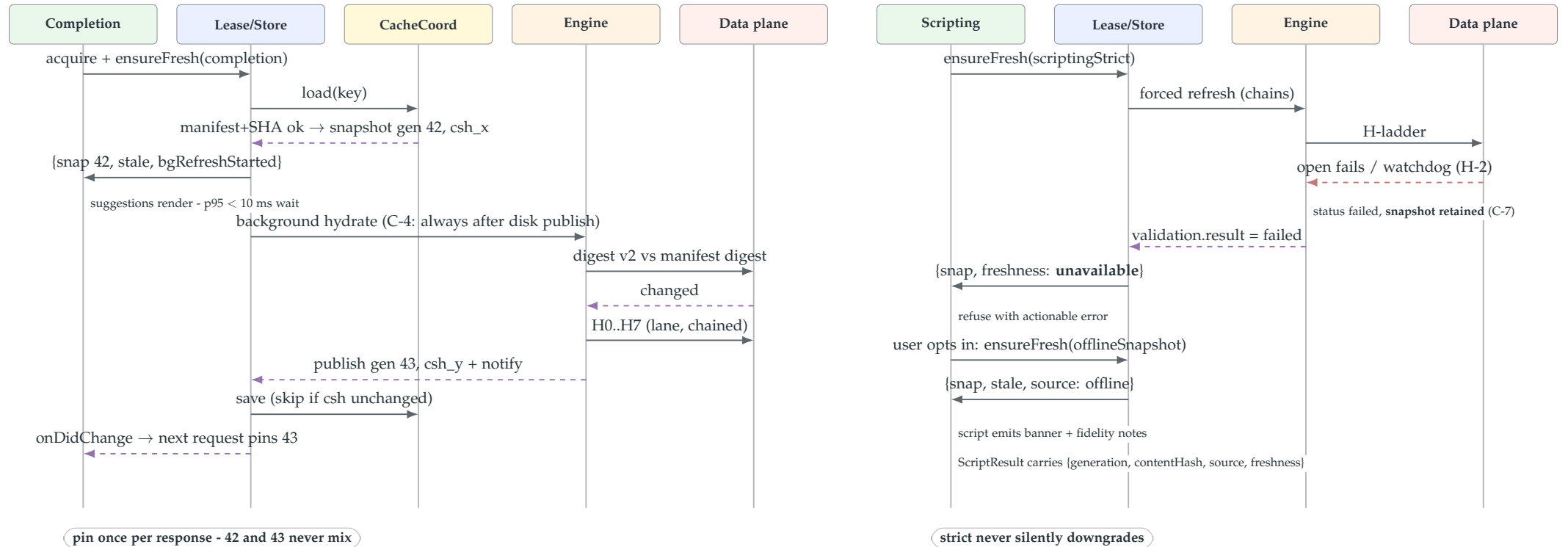
A5. Two sequences: cold-start allowStale, and requireLive with a refusal leg

Left: the headline win - completion answers from disk in milliseconds while truth catches up. Right: strict scripting never silently downgrades.

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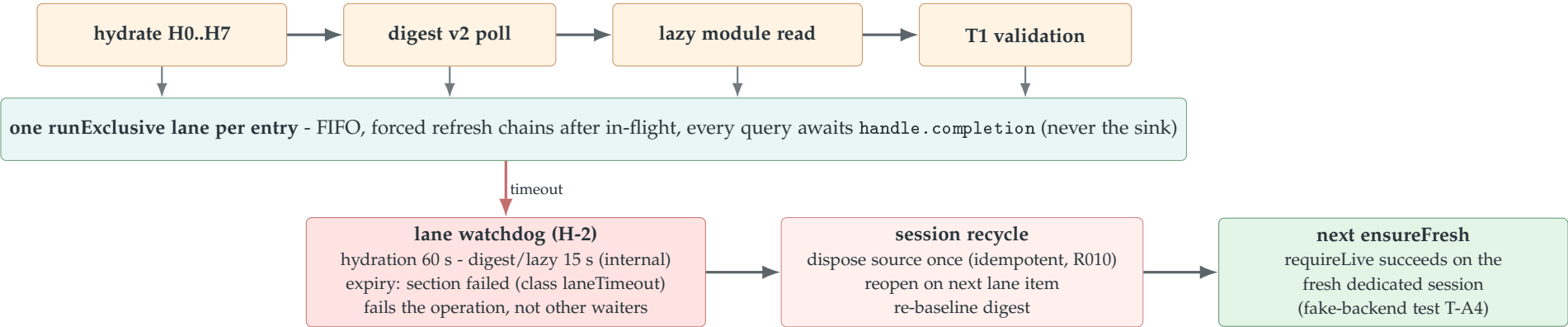
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A6. The session lane under failure: watchdog, recycle, and the STS2 interlocks

The completion-reaction discipline assumes exactly one `v2/query.complete` per accepted `execute`. Until STS2-R008/R009/R010 land, the lane must be able to survive a completion that never comes.

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service-side dependencies (STS2 review)
R008: `query.dispose` can suppress `query.complete`
R009: `dispose` releases the connection before the pump stops
R010: duplicate close can orphan the first waiter
Extension rule until fixed: **never dispose an awaited lane query**;
cancellation = stop waiting (C-9) + watchdog recycle only

every await point here has been a race at least once

Determinism marker (C-2). Generations are process-scoped integers; two windows can both mint “43” with different content. The cross-process key for prompt caching, replay comparison, and feature capture is `contentHash` (canonical payload SHA-256). It rides result metadata and spans - **never prompt bytes**. Live-hydrated and cache-rehydrated snapshots with identical data hash identically by construction, which is also the codec round-trip proof.

dispose during hydrate: bounded drain, then hard-dispose